

CONNECTION DESIGN ADVISOR

Project: Quick Analysis

Generated: 2026-04-24T19:12:42.136758+00:00

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Greetings. I am STRUCTMIND AI, your Connection Design Advisor.

Designing connections for a mixed SMF (Special Moment Frame) and OCBF (Ordinary Concentrically Braced Frame) system requires careful adherence to both AISC 360-22 Specification for Structural Steel Buildings and AISC 341-22 Seismic Provisions for Structural Steel Buildings. The "optimal" connection is a balance of seismic performance, fabrication economy, and erection speed.

Below are my recommendations for the key connections in your 3-storey structure.

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1. SMF Beam-to-Column Connections (Interior & Exterior Joints)

These connections form the primary lateral system in the moment frame lines and must be designed for high ductility and inelastic rotation.

- 1. Recommended Connection Type: Bolted Unstiffened End Plate (BU EP)
- This is a prequalified moment connection per AISC 341-22, Chapter E and detailed in AISC 358-22. It is highly efficient for both fabrication (shop welding) and erection (field bolting).
- 2. Governing Limit States (AISC 360-22 & 341-22):
 - The connection design is governed by capacity design principles, where the connection must be stronger than the hinging beam. The required moment strength is based on the probable plastic moment of the beam, $M_{pr} = C_{pr} R_y F_y \cdot Z_x$.
 - Beam-Side:
 - End Plate Flexural Yielding (AISC 360, J10.8)
 - Bolt Tension Rupture (AISC 360, J3.6)
 - Weld Strength: CJP groove weld of beam flanges to end plate; fillet or CJP weld of beam web to end plate. (AISC 360, J2)
 - Column-Side:
 - Column Panel Zone Shear Yielding (AISC 341, E3.6e). This is often the governing limit state for the column, potentially requiring doubler plates.

- Column Flange Local Bending due to prying action. (AISC 360, J10)
- Column Web Local Yielding / Crippling at beam flange locations. (AISC 360, J10.2 & J10.3). May require column stiffeners (continuity plates).
- Bolt Shear Rupture for the shear transfer portion of the bolt group. (AISC 360, J3.6)
- 3. Bolt / Weld Sizing Guidance:
 - Bolts: Use high-strength ASTM F3125 Grade A325 or A490 (or F1852/F2280 TC bolts). Bolts are pretensioned and designed for combined tension (from moment) and shear. Bolt diameter and quantity are dictated by the formulas in AISC 358 to ensure strength and prevent premature failure modes.
 - Welds: The beam-to-end-plate welds are typically CJP groove welds at the flanges to fully develop the flange force. The web weld is sized to transfer the required shear.
- 4. Constructability Score: 8/10
 - Remarks: Excellent choice. The end plate is welded to the beam in the shop under controlled conditions. Erection is fast and simple, requiring only field bolting. Reduces need for field welding and complex NDT, improving project schedule and quality control.
- 5. Alternate Solutions:
 - Reduced Beam Section (RBS): A classic prequalified SMF connection. Involves creating a "dog-bone" cut in the beam flanges to force the plastic hinge away from the column face. Requires more complex CNC cutting in the shop but is a very reliable detail.
 - Bolted Flange Plate (BFP): Another prequalified bolted option. Plates are bolted to the top and bottom beam flanges and welded or bolted to the column. Can be more labor-intensive than an end plate.

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2. OCBF Brace-to-Gusset & Gusset-to-Frame Connection

This connection must accommodate brace buckling while ensuring the load path is maintained. Design is governed by AISC 341-22, Chapter F.

- 1. Recommended Connection Type: Corner Gusset Plate
 - A gusset plate is shop-welded to the beam and/or column and field-bolted to the brace (typically a Wide Flange or HSS member).
- 2. Governing Limit States (AISC 360-22 & 341-22):
 - The connection is designed for the expected strength of the brace.
 - In tension: $R_y F_y A_g$ (Tensile Yielding of Brace)
 - In compression: $1.1 R_y P_n$ (Expected Buckling Strength) or the maximum force the system can deliver. OCBF connections are not required to be designed for the post-buckling capacity of the brace.
- Brace-to-Gusset:

- Bolt Shear and Bearing on brace and gusset. (AISC 360, J3.6 & J3.10)
- Brace Member Checks: Tensile Yielding/Rupture, Block Shear Rupture. (AISC 360, J4)
- Gusset Plate:
 - Gusset Plate Tensile Yielding on the Whitmore effective width. (AISC 360, J4.1)
 - Gusset Plate Buckling (compressive strength). (AISC 360, J4.4)
 - Block Shear Rupture at the bolt group. (AISC 360, J4.3)
- Gusset-to-Frame:
 - Weld Strength of gusset to beam/column flanges/web. (AISC 360, J2)
 - Beam/Column Member Checks for forces imparted by the gusset.
- 3. Bolt / Weld Sizing Guidance:
 - Bolts: Use high-strength, slip-critical bolts (Group B). Bearing-type connections are permitted if slip is not a concern, but SC is preferred for seismic systems.
 - Welds: Size fillet welds to develop the required strength at the gusset-to-frame interface.
 - Gusset: Thickness is often governed by buckling. Ensure a clearance of at least $2 * t_{plate}$ is provided between the end of the brace and the frame members to allow for brace end rotation during buckling.
- 4. Constructability Score: 7/10
 - Remarks: A very standard and well-understood connection. Fabrication is straightforward. Fit-up in the field can be challenging if tolerances are not maintained. Shop-welding the gusset to the column is often preferred for erection sequencing.
- 5. Alternate Solutions:
 - End-Plate Gusset for HSS Braces: An end plate is welded to the HSS brace and then bolted directly to the gusset. Can simplify the gusset geometry.

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3. Column Base Plates

This connection anchors the entire lateral force-resisting system to the foundation.

- 1. Recommended Connection Type: Exposed Column Base Plate
 - For both SMF and OCBF columns, a fixed base is typically assumed and designed to transfer axial load, shear, and high overturning moments.
- 2. Governing Limit States (AISC 360-22, J8 & ACI 318-19, Ch. 17):
 - Design is based on amplified seismic loads or capacity-limited forces from the yielding frame above.
 - Bearing on Concrete: Concrete Crushing. (AISC 360, J8)

- Base Plate Flexural Yielding: The plate itself bending under the tension/compression couple.
- Anchor Rods:
- Steel Strength in Tension / Shear. (ACI 318, Ch. 17)
- Concrete Breakout Strength (Tension and Shear). This is frequently the governing limit state for anchors.
- Concrete Pullout Strength.
- Concrete Side-Face Blowout.
- Column-to-Base Plate Weld: Weld strength must be sufficient to transfer all forces from the column. (AISC 360, J2)
- Shear Transfer: Shear can be transferred via anchor rod shear, a shear lug, or friction (if sufficient compression is present).
- 3. Bolt / Weld Sizing Guidance:
- Anchor Rods: Use high-strength, ductile steel (e.g., ASTM F1554 Grade 55). Size diameter and embedment length to resist combined tension and shear, paying close attention to concrete breakout.
- Base Plate: Thickness is determined by the flexural demand from bearing pressures. A thicker plate reduces prying on anchor bolts.
- Welds: Often CJP or large fillet welds are required to develop the column section capacity, especially for SMF columns.
- 4. Constructability Score: 8/10
- Remarks: Standard detail, but anchor rod placement is critical. Using a survey-verified template is mandatory. Double nuts (leveling nut and top nut) are standard practice for setting plate elevation and ensuring proper bearing after grouting.
- 5. Alternate Solutions:
- Embedded Column Base: The column is embedded directly into the concrete foundation. Provides excellent fixity but is more complex to construct and detail.
- Proprietary Grouting Systems: High-strength, non-shrink grouts can enhance performance but add cost.

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4. Splice Connections (Column & Beam)

- 1. Recommended Connection Type:
- Column Splice: Bolted Flange Plate Splice.
- Beam Splice (Collector/Gravity): Bolted Web and/or Flange Plate Splice.
- 2. Governing Limit States (AISC 360-22, J6 & 341-22, D2.5):

- Column Splice (SMF/OCBF): Per AISC 341-22, splices must be located away from plastic hinge zones (typically ~4ft above the floor). They must be designed to develop:
 - The required shear strength from load combinations.
 - A moment strength of at least $R_y F_y Z_x$ of the smaller column connected.
 - Limit states include Bolt Shear/Bearing, Plate Yielding/Rupture, and Block Shear.
- Beam Splice (for collectors or non-LFRS beams): Located at points of low moment.
 - Designed for the required shear and moment at the splice location.
 - Limit states are standard checks: Bolt Shear/Bearing, Plate Yielding/Rupture, Block Shear, and Flexural Rupture of the Coped Section if applicable.
- 3. Bolt / Weld Sizing Guidance:
 - Bolts: Use slip-critical bolts for all seismic-system splices. A325-N bolts may be sufficient for simple gravity beam splices.
 - Plates: Plate thickness and width are sized to provide the required area and section modulus to transfer the design forces.
- 4. Constructability Score: 9/10
 - Remarks: Bolted splices are highly constructible and the industry standard. They allow for shipping members in manageable lengths and facilitate safe, rapid erection. Column splices are typically staggered between frame lines.
- 5. Alternate Solutions:
 - Complete Joint Penetration (CJP) Welded Splice: Provides a seamless load path but is very expensive due to the need for highly skilled field welders, access platforms, and extensive non-destructive testing (NDT). Generally avoided unless absolutely necessary for architectural or structural reasons.